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The Need for a Geohazard App in Herkimer, NY: Government Entity Preparedness for Future Flooding

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Introduction

Herkimer, New York experienced a major flood in July of 2013 which required the assistance of the Federal Emergency Management Administration and the Army Corp of Engineers (Eisenstadt, 2013). According to some respondents interviewed for this study, although the flood was seen as a "one hundred year event" due to heavy rains, the village is situated in a location that increases the risk of another flood should the West Canada River or Hinkley Reservoir breach the levees in place. (Figure 1.) We conducted an anonymous and non-recorded survey of people working in government entities in the town of Herkimer, NY to ascertain the need for a dynamic mobile application that can provide real-time flood data, including evacuation routes and the ability to monitor flood damage. Previous research has shown that flood simulation and reconstruction of a flood event with capabilities of damage evaluation is possible (Swan et al., 2016). Utilizing this technology in a mobile app would enhance the resilience of the town's infrastructure and quick recovery from a flood event.

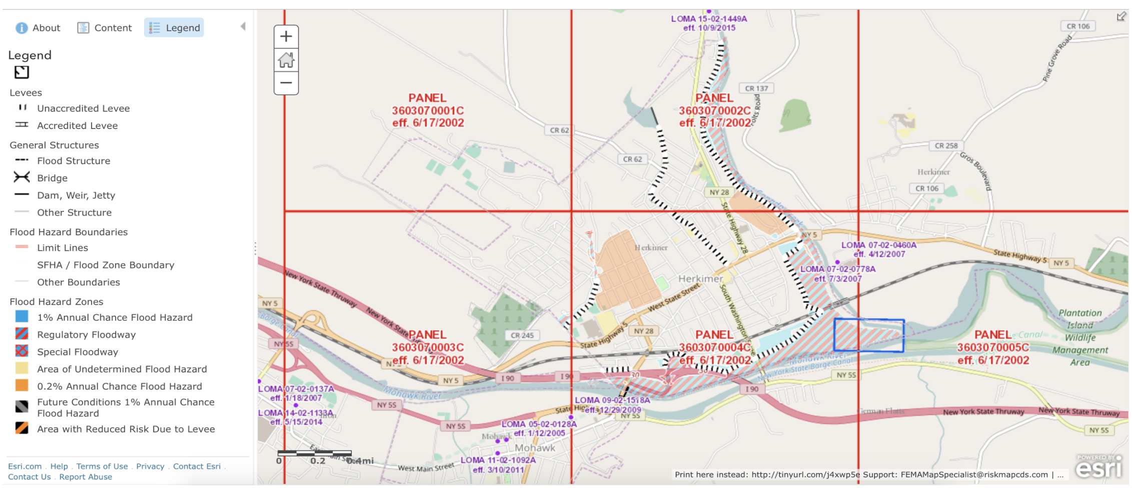

Figure 1: FEMA Surveyed Area of Flood Vulnerability of the study area

Methods

We conducted nineteen surveys of government institutions serving Herkimer, NY. Herkimer was selected as a location for this study because the area had experienced a major flood event in 2013. Additionally, we used this location as an example as results from interviews in this region may be applied to other regions susceptible to a flood hazard. The institutions included the town hall, post office, fire department, emergency services, police department, highway department, transit authority, mental health care facility, health care facility, and the zoning office, among others. The questionnaire used for the interviews was comprised of ten questions that asked respondents to report how flooding in the past has affected their operations, and how the impact of future flood might be mitigated by information provided by a geohazard app that can provide real time data. Examples of questions included: "how likely do you think it is that there will be a flood in the future?," "Would you be interested in a flood app that can show the extent of flood damage?," "Would you be interested in a flood app that can connect you with emergency responders in the event of a flood?," and "Do you think a flood app would increase the speed of recovery after a flood?"

A statistical software package (SPSS) was used to analyze the responses of the surveys. In the initial analysis of the first round of results Chi-Square, Phi and Cramer's V tests were employed to determine whether there were relationships and how strong is the relationship between the answers to different questions on the survey. After these analyses questions were grouped into those related to respondents' experiences with flooding in the past (questions 1-4; e.g., amount of damage, probability of future flooding) and the need for the geohazard app (questions 5-8; e.g., interest in a flood app that would connect users to emergency responders, provide real time evacuation routes, speed recovery, and provide daily groundwater levels). Scores were computed for each group and then correlated with Pearson's product moment to determine whether there was a relationship between respondents' experiences with previous flooding and the need for the mobile app. Higher scores on each grouping of questions indicated a greater need for the app.

Results

Descriptive statistics on the 19 responses has shown the mean, median, minimum, maximum, standard deviation and range values. The means and standard deviations are reported in Table 1. The range of values was from 1 to 5, and the median value was 3.1. Chi-square analyses of the 19 responses did not reveal any significant relationships between any questions on the survey.

Table 1: Descriptive statistics for each question

Question #	Mean	SD
Q1	2.05	1.2241
Q2	2.53	1.073
Q3	4.05	1.353
Q4	2.53	1.611
Q5	4.37	1.012
Q6	2.95	1.649
Q7	3.58	1.427
Q8	4	1
Q9	1.89	1.286
Q10	3.47	1.073

The next analysis included the recording to different variables (grouping) of scores to represent experiences with flooding and the expressed need for the app. The range of scores for group 1 questions was from 4-18, while group 2 scores ranged from 10-29. The mean score for group 1 questions (experiences with flooding) was 11.15, while the mean score for group 2 questions (need for the app) had a mean of 20.26. The results of the correlation between these two groups of questions was a value of $r = 0.37$ which was significant at $p < 0.05$, indicating a moderate positive relationship between experience with flooding and the need for the app. Other analysis showed the relationships between those in the survey that had answered one-three or from four-five to see if there was a direct relationship between the answers, or if there was a wide range. In addition, a second correlation graph was created on the individual responses to see if any questions may have a trend. Ultimately, no strong correlation could be determined due to the small selection pool of twenty. The only trend attained in the data is in government municipals - those that suffered from flooding often were more likely to positively respond to the app questions while those who did not respond positively were not affected more than once. Below is the chart showing the range in responses, and the correlations between high and low values on the scores. (Fig. 2 and 3.)

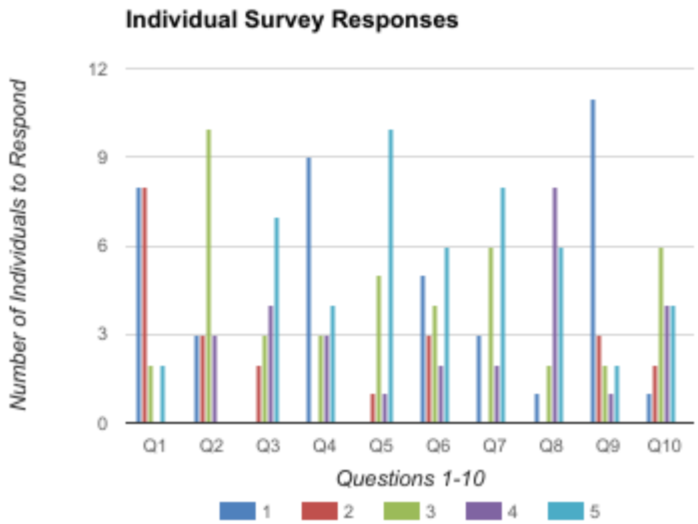

Figure 2: Bar Graph Showing Correlation Between Individual Answers to Survey

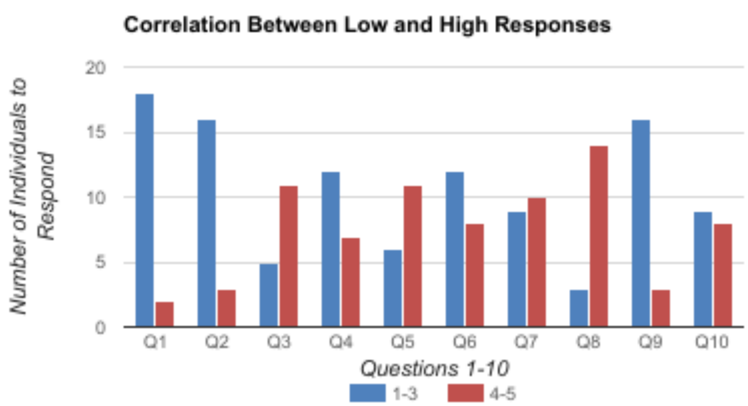

Figure 3: Correlation Between Grouped Responses High vs. Low Values

Several respondents noted that the county has measures in place to monitor flooding and would likely provide more collaborative real-time data than a geohazard app, depending upon the data used to construct the application. According to respondents who work for the town, the county is involved in a flood mitigation study. There were differences in the perceived need for an app between government entities. For example, some government officials believed that the precautions put in place in the town would serve as a better alert to residents than an app because the town services such as emergency services and the fire department have more nuanced approaches to communications. However, other respondents did not report any concern for future flooding, but did indicate that residents may find a need for or be interested in this app if there were flooding in the future. Below is the flood map figure provided by FEMA. Despite the vulnerability in the region, government entities did not suffer themselves in the flood - but rather as a collective whole suffered from the damages.

Discussion

The results of the survey used in this study has shown that despite claims that there may be no future flooding, there may be an interest in a geohazard app. Although chi-square analysis did not reveal significant relationships between questions answered by respondents, when grouping the answers to questions by “experiences with flooding” and “perceived need for the app” there was a significant relationship measured by the Cramer’s V value. Comments provided by respondents indicated that while a future flood is unlikely, a geohazard app may be useful for constituents. According to the government employees interviewed it is possible that residents have a different perception of need for the app Additional research on resident responses to this survey is needed to substantiate these views. Furthermore, our survey has shown that a better understanding of a professional geohazard mobile app by the government entities or by the residents would facilitate to understand the differences between the type of information the app can provide and the type of information the town readily possesses.

A common note amongst the government municipals was the lack of funding that the town has. Being in close proximity to many others towns also affected by flooding, group collaboration is essential in planning, managing, and mitigating all effects of flooding. With the recent flood event in 2013, the towns and villages worked closely together with the aid of the federal government, but after the immediate cleanup of the flood a major need for infrastructure changes arose. Funding at this time has not come forward from other towns due to financial strains, which leaves the town of Herkimer County vulnerable to future flood events. The different entities all commented to say that it is important for close collaboration between all of the different departments and that future flood event management relies on this heavily. By increasing our pool of responders, we may be able to show a better statistical correlation in the Cramer's v values that would give a more fundamentally sound answer to whether or not the government municipals in Herkimer County would respond to a real-time app.

Conclusion

Future research should compare the responses of government entities and those of residents and business as they all have different needs when it comes to flooding in the area. In addition, the amount of subjects should be greater than twenty in order to have more significant results when trying to evaluate the Cramer’s V and Chi-square values. Despite the claims by most respondents that flooding in the future is unlikely, government officials in charge of all of the towns and the entirety of the county were quite positive that flooding will continue to be a problem within the Herkimer County.

Acknowledgement

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References

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Poster Presentation